Discovering the Ornstein–Uhlenbeck Fokker–Planck Equation from Density Data

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Code and notebooks

Motivation

Physical laws are often derived from first principles. When active terms are unknown, **data-driven equation discovery** offers a complementary route: infer a sparse governing equation directly from observations.

Research question and benchmark

Can sparse regression recover a Fokker–Planck equation directly from density data, including noisy trajectories?

Our benchmark is the Ornstein–Uhlenbeck process and its conservative density equation:

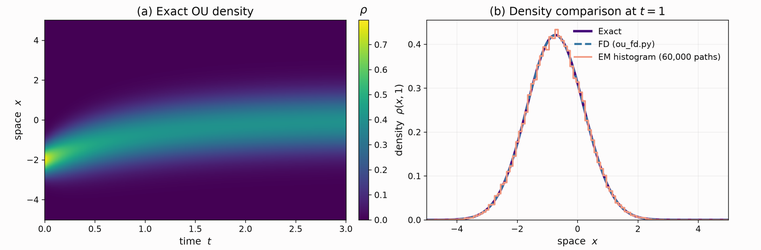
$$\mathrm{d}X_t = -\kappa X_t\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}W_t, \qquad \boxed{\rho_t = \kappa\,\partial_x(x\rho) + D\rho_{xx}}.$$

With $\kappa = D = 1$, $(x,t) \in [-5,5]\times[0,3]$ and $X_0 \sim \mathcal{N}(-2, 0.5^2)$, the true support is $\{\partial_x(x\rho), \rho_{xx}\}$ with coefficients $(1,1)$.

Because both the dynamics and density are known, departures from $(1,1)$ isolate errors due to data generation, numerical differentiation, library design or sparse regression.

Density data used

Aspect	Exact	Finite difference	Euler–Maruyama
Construction	analytic OU density	numerical FPE solution	60,000 paths + histogram
Main uncertainty	none (reference)	space–time discretisation	time stepping, sampling and binning



The three sources agree visually, but small bin-to-bin fluctuations in the EM histogram are strongly magnified by pointwise numerical derivatives.

Sparse identification of the PDE

Ordinary SINDy solves a sparse regression problem,

$$\mathbf{b} \approx \Theta\boldsymbol{\xi}, \qquad \mathbf{b} = \mathrm{vec}(\rho_t),$$

using sequentially thresholded least squares (STLSQ). Each column of Θ represents a candidate PDE term; a representative strong-form library is

$$\rho,\ \rho_x,\ \partial_x(x\rho),\ \partial_x(x^2\rho),\ \rho_{xx},\ \rho_{xxx}, \ldots$$

Noise: if $\widehat{\rho} = \rho + \varepsilon$, finite differences amplify noise approximately as ε/h for first derivatives and ε/h^2 for second derivatives.

Identifiability: larger libraries are not automatically better. The identity $\partial_x(x\rho) = \rho + x\rho_x$ creates dependent columns and non-unique coefficients. Prefer a compact conservative library; otherwise compare algebraically equivalent PDEs.

Weak SINDy: differentiate the test function, not the data

On each randomly sampled space–time subdomain Ω_k, integrate against a smooth compactly supported test function ϕ_k and transfer derivatives by parts:

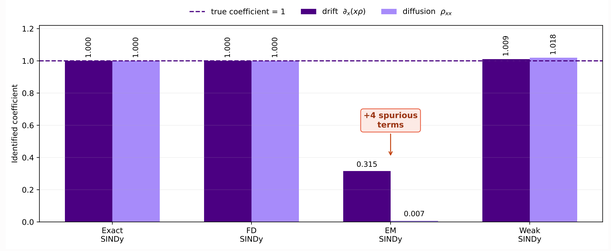
$$\int_{\Omega_k}\rho_t\phi_k = -\int_{\Omega_k}\rho\,(\phi_k)_t, \qquad \int_{\Omega_k} q_x\phi_k = -\int_{\Omega_k} q\,(\phi_k)_x,$$
$$\int_{\Omega_k}\rho_{xx}\phi_k = \int_{\Omega_k}\rho\,(\phi_k)_{xx}.$$

Boundary terms vanish by construction. The observed density appears only inside local weighted integrals, so noise is averaged rather than differentiated.

Library 1	five selected columns: $\partial_x(x\rho)$, $\partial_x(x^2\rho)$, $\partial_x(x\rho^2)$, $\partial_x(x^2\rho^2)$, ρ_{xx}
Library 2	all 15 columns: five fields with derivative orders $0,1,2$

Each subdomain supplies one weak equation: K sets the number of equations, while width sets the local averaging scale.

Weak integration restores coefficient recovery



SINDy + Exact/FD
correct support
coefficients close to 1

SINDy + EM
diffusion nearly lost
four spurious terms

Weak SINDy + EM
correct support
coefficients close to 1

At $K = 500$ and 25% requested full width, both weak libraries recover the correct support with $\widehat{\kappa} = 1.009$ and $\widehat{D} = 1.018$.

K–width sensitivity across weak-subdomain seeds

On the fixed EM density, we sweep $K = 100, 250, 500, 750$ and requested full widths $10\%, 25\%, 50\%, 75\%$ using seeds 319, 801 and 2002. Cells report correct-support count out of three and median coefficient error; stars mark the camera-error setting.

25% is the safest common choice: both libraries recover the true support for every K and seed. At 75%, Library 2 fails in all three seeds despite a median residual of only 5.3×10^{-4}, so a small residual does not certify the correct PDE.

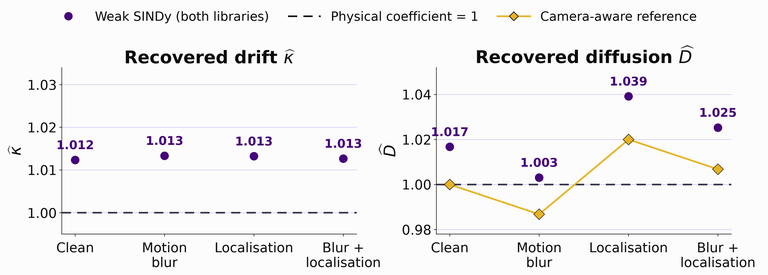


Robustness to camera errors

The same 60,000 OU paths are re-observed under four camera conditions:

Clean	midpoint positions
Motion blur	exposure average $X^{\mathrm{blur}}_{m,\alpha} = \frac{1}{\Delta}\int_{t_\alpha}^{t_\alpha+\Delta} X_m(s)\,ds$, $\Delta = 0.04$
Localisation	additive Gaussian noise $X^{\mathrm{loc}} = X^{\mathrm{clean}} + \epsilon$ with $\epsilon \sim N(0, \sigma^2_{\mathrm{loc}})$
Both	motion blur + localisation

Fit: 60,000 paths; $K = 500$; 25% width; both libraries.



Across all conditions, both libraries recover exactly $\partial_x(x\rho)$ and ρ_{xx}; camera-aware coefficient error is 1.47%–1.63%.

Camera bias remains: camera errors leave the selected PDE terms unchanged but shift the effective diffusion coefficient. Weak SINDy reduces derivative-amplified noise; it does not correct the camera observation model.

References

[1] S. L. Brunton, J. L. Proctor and J. N. Kutz. Sparse identification of nonlinear dynamics. *PNAS*, 113:3932–3937, 2016.

[2] S. H. Rudy, S. L. Brunton, J. L. Proctor and J. N. Kutz. Data-driven discovery of PDEs. *Science Advances*, 3:e1602614, 2017.

[3] D. A. Messenger and D. M. Bortz. Weak SINDy for PDEs. *J. Computational Physics*, 443:110525, 2021.

[4] A. J. Berglund. Statistics of camera-based single-particle tracking. *Physical Review E*, 82:011917, 2010.